

CsharpDay05Assignment

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- Note: Part01 and Part02 problems in the resportory
- Part01 questions:

Questions:-

1. Purpose of the finally block
 - Executes always, whether an exception occurs or not.
 - Used for cleanup tasks like closing files or printing messages.
2. How `int.TryParse()` improves program robustness compared to `int.Parse()`
 - `int.Parse()` throws an exception on invalid input.
 - `int.TryParse()` returns true/false, allowing safe handling without crashing.
3. What exception occurs when trying to access Value on a null `Nullable<T>`?
 - Throws `InvalidOperationException` if the value is null.
4. Why is it necessary to check array bounds before accessing elements?
 - Prevents `IndexOutOfRangeException`.
 - Ensures program stability and prevents crashes.
5. How is the `GetLength(dimension)` method used in multi-dimensional arrays?
 - Returns the size of a specific dimension.
 - Example: `rows = array.GetLength(0)`, `cols = array.GetLength(1)`
6. How does the memory allocation differ between jagged arrays and rectangular arrays?
 - Rectangular arrays: single memory block, all rows same length.
 - Jagged arrays: array of arrays, each row allocated separately, lengths can vary.

7. What is the purpose of nullable reference types in C#?
 - Allows references (like string?) to explicitly be null.
 - Helps catch null reference errors at compile time.
 - Null-forgiveness operator (!) suppresses warnings when you are sure it's not null.
8. What is the performance impact of boxing and unboxing in C#?
 - Boxing/unboxing causes heap allocation and runtime checks.
 - Frequent use reduces performance.
9. Why must out parameters be initialized inside the method?
 - out is uninitialized when passed in.
 - Method must assign a value before returning.
10. Why must optional parameters always appear at the end of a method's parameter list?
 - Arguments are matched left to right.
 - Placing optional parameters last avoids ambiguity.
11. How does the null propagation operator prevent `NullReferenceException`?
 - Accesses members only if object is not null.
 - Returns null instead of throwing an exception.
 - Example: `length = nullableArray?.Length`
12. When is a switch expression preferred over a traditional if statement?
 - Concise, readable, ideal for mapping specific values.
 - If statements are better for ranges or complex conditions.
13. What are the limitations of the `params` keyword in method definitions?
 - Only one `params` parameter per method.
 - Must be last parameter.
 - Cannot have optional parameters after a `params` parameter.